JAVA · 1D ARRAYS

Creating Arrays

Declare it. Build it. Fill it.



OBJECTIVES

Today's goals

1

Declare and create an array with **new**

2

Create an array with an **initializer list**

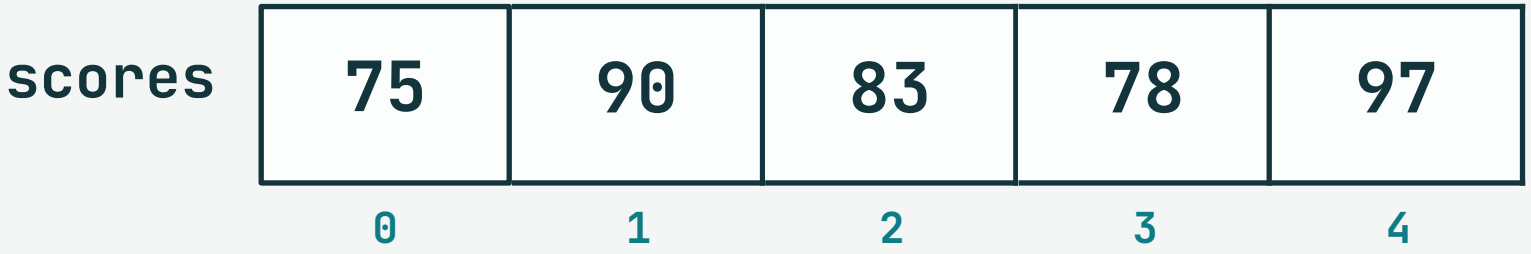
3

Find an array's **length**

THE IDEA

What is an array?

One variable that holds **many values** of the **same type**.



same type | fixed size | index starts at 0

THE IDEA

Like a list, with rules

You have used lists before. A Java array is stricter.

✓

One type

An int array holds only ints.

Fixed size

It can’t grow or shrink.

Typed

You state the type when you create it.

STEP 1

Declare the array

```
int[] scores;
```

`int`

type of each element

`[ ]`

makes it an array

`scores`

the variable name

No array exists yet. **scores** holds **null**.

STEP 2

Create it with new

```
scores = new int[5];
```

`new`

builds a new array object

`int`

element type, must match

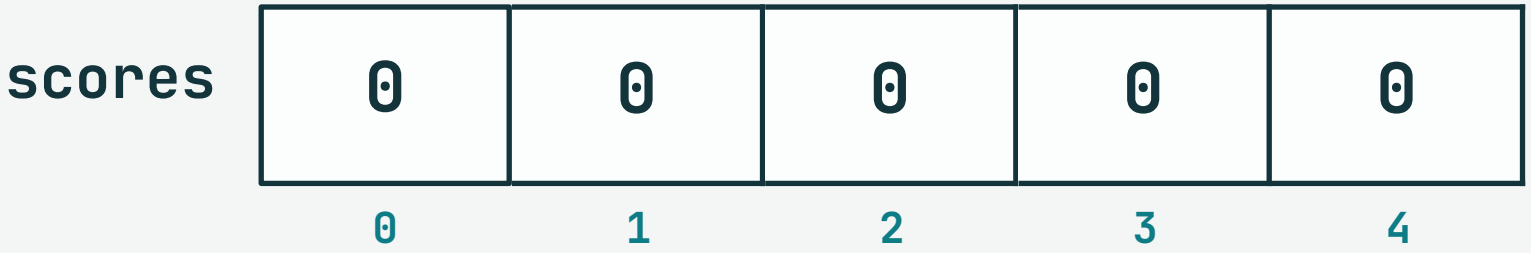
`[5]`

number of elements

SHORTCUT

Both steps in one line

```
int[] scores = new int[5];
```



Every element starts at the default value: **0**.

STEP 2

Default values

Type	Default	Example
`int`	`0`	`new int[3]`
`double`	`0.0`	`new double[3]`
`boolean`	`false`	`new boolean[3]`
`String`	`null`	`new String[3]`

A String array starts with **null**: no String object yet.

YOUR TURN

What's inside?

```
double[] prices = new double[4];
```

Draw the boxes. Write each value and its index.



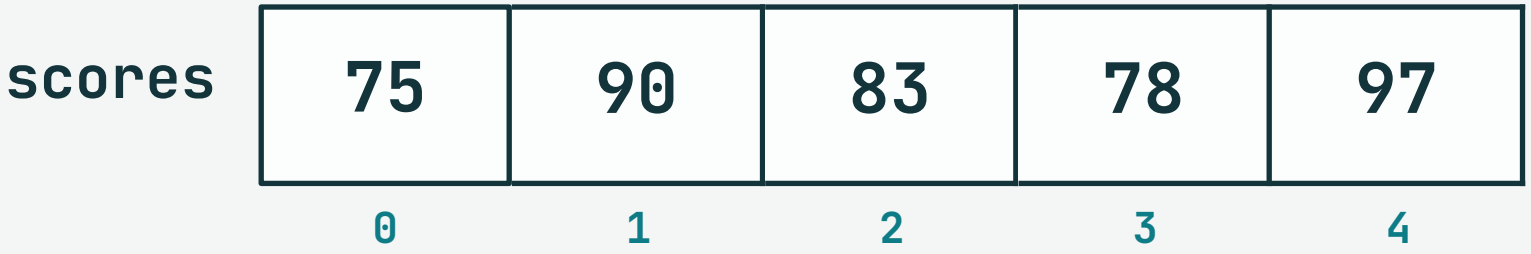
Students, write your response!

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ANOTHER WAY

Initializer list

```
int[] scores = {75, 90, 83, 78, 97};
```



Java counts the values for you. Size: **5**.

ANOTHER WAY

Initializer list rules

```
double[] temps = {98.6, 101.2};
```

works

Every value is a double.

```
int[] ages = {15, "16", 17};
```

error

Every value must match the array's type.

```
int[] a; a = {1, 2, 3};
```

error

{ } alone only works on the declaration line.

CHOOSING

new or { } ?

Know the values? Use { }

```
int[] dice = {1, 2, 3, 4, 5, 6};
```

Know only the size? Use new

```
String[] names = new String[30];
```

SIZE

How long is it?

```
int[] scores = {75, 90, 83, 78, 97};
System.out.println(scores.length);
```

Console

5

length has no ( )

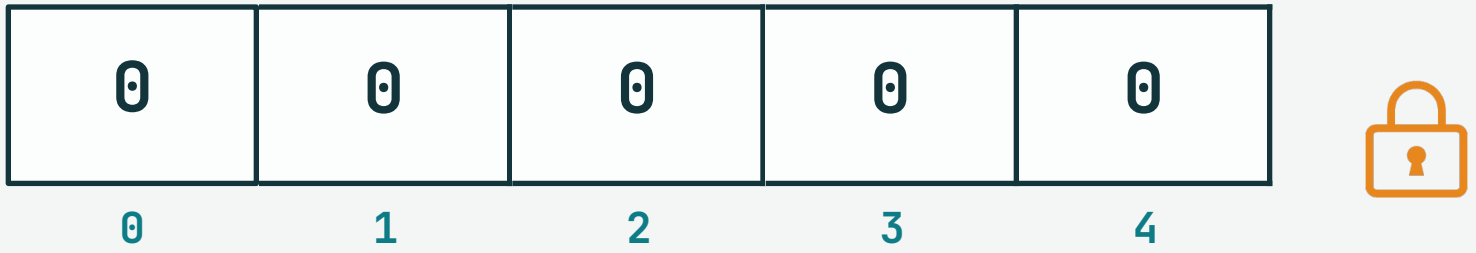
Last index = **length – 1**

length never changes

SIZE

Arrays can’t grow

```
int[] scores = new int[5];
```



Need 7 scores? Create a **new** array of size 7.

YOUR TURN

Fix each line

#	Code
1	`int[] a = new int(5);`
2	`int[] a = new double[5];`
3	`int[] a = [1, 2, 3];`
4	`int n = a.length();`
5	`int[] a = new int[];`



Students, draw anywhere on this slide!

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YOUR TURN

Write the code

1. An array named **nums** that holds 10 doubles.

2. An array named **colors** holding "red", "green", "blue".

3. What are the first and last index of **colors**?



Students, draw anywhere on this slide!

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CHECK

Which line makes an array of 7 Strings?

A `String[] a = new String[7];`

B `String[] a = String[7];`

C `String a = new String[7];`

D `String[] a = new String(7);`

PRACTICE

Now on CodeHS

</> Open **[CodeHS lesson 4.3** and complete the exercises.

WRAP UP

Key vocabulary

Term	Meaning
array	one variable holding many values of one type
element	a single value stored in an array
index	an element’s position, starting at 0
length	how many elements the array holds
initializer list	values inside { } that create an array
default value	what new fills each element with